Supplementary material for “Evaluating Authorization Regimes and Runtime Guardrail Configurations for Tool-Using LLM Agents”

These twenty tables and five historical figures provide detailed results referenced by the main manuscript. They introduce no additional hypothesis family or claim beyond the analyses reported in the article.

Table S1: Online-state source audit

Conclusion	Audited units	Independently reviewed	Claim use
No online gold observed on covered path	9	0	Path-scoped only
Not a runtime guard	10	0	Taxonomy only
Mixed or path dependent	1	0	Path-scoped caution
Inconclusive	2	0	No substantive claim

Table S2: Completed medium security and utility outcomes (part A)

Row	Condition	Unsafe effects / 36 attacks	Unsafe-effect rate (Wilson 95% CI)	Tool completions / 24
1	No defense	8/36	22.22% (11.72–38.08)	12/24
2	Safe prompt / spotlighting	8/36	22.22% (11.72–38.08)	11/24
3	PromptGuard (detector-only)	8/36	22.22% (11.72–38.08)	12/24
4	Structural baseline	8/36	22.22% (11.72–38.08)	12/24
5	Policy package (m3_policy_only)	0/36	0.00% (0.00–9.64)	7/24
6	Full package (m3_full)	0/36	0.00% (0.00–9.64)	6/24

Table S2: Completed medium security and utility outcomes (part B)

Row	Condition	Unsafe effects / 36 attacks	Tool TCR (Wilson 95% CI)	No-tool TCR
1	No defense	8/36	50.00% (31.43–68.57)	12/12
2	Safe prompt / spotlighting	8/36	45.83% (27.89–64.93)	12/12
3	PromptGuard (detector-only)	8/36	50.00% (31.43–68.57)	12/12
4	Structural baseline	8/36	50.00% (31.43–68.57)	12/12
5	Policy package (m3_policy_only)	0/36	29.17% (14.91–49.17)	12/12
6	Full package (m3_full)	0/36	25.00% (12.00–44.90)	12/12

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S3: Prespecified paired comparisons

Comparison	Treatment minus baseline	Exact McNemar p	Holm-adjusted p	10,000-cluster bootstrap 95% CI
Full package vs structural, unsafe effect	−22.22 pp	0.0078125	0.03125	−36.11 to −8.33 pp
Full vs policy package, unsafe effect	0.00 pp	1.0	1.0	0.00 to 0.00 pp
Full package vs structural, tool TCR	−25.00 pp	0.03125	0.09375	−41.67 to −8.33 pp
Full vs policy package, tool TCR	−4.17 pp	1.0	1.0	−12.50 to 0.00 pp

Table S4: Runtime burden and operational quality (part A)

Row	Condition	ASK / 72	Benign ASK rate	Operational failures / 72
1	No defense	0	0.00%	4
2	Safe prompt / spotlighting	0	0.00%	5
3	PromptGuard (detector-only)	0	0.00%	5
4	Structural baseline	0	0.00%	4
5	Policy package	16	19.44%	1

Table S4: Runtime burden and operational quality (part A) (continued)

Row	Condition	ASK / 72	Benign ASK rate	Operational failures / 72
6	Full package	22	22.22%	2

Table S4: Runtime burden and operational quality (part B)

Row	Condition	ASK / 72	Provider calls	Trace-bundle cost (CNY)
1	No defense	0	181	0.1821176
2	Safe prompt / spotlighting	0	167	0.1812942
3	PromptGuard (detector-only)	0	180	0.0986064
4	Structural baseline	0	249	0.2568822
5	Policy package	16	112	0.0548220
6	Full package	22	117	0.1441772

Table S4: Runtime burden and operational quality (part C)

Row	Condition	ASK / 72	Total latency p95
1	No defense	0	16.924 s
2	Safe prompt / spotlighting	0	15.971 s
3	PromptGuard (detector-only)	0	18.249 s
4	Structural baseline	0	21.251 s
5	Policy package	16	7.897 s
6	Full package	22	10.701 s

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S5: Public-primary benchmark strata

Benchmark / endpoint	Structural baseline	Policy package	Full package
AgentDojo attacks, unsafe effect	87/300 (29.00%)	13/300 (4.33%)	29/300 (9.67%)
M3-v2 attacks, unsafe effect	46/300 (15.33%)	0/300 (0.00%)	9/300 (3.00%)
AgentDojo benign tool, TCR	67/97 (69.07%)	13/97 (13.40%)	13/97 (13.40%)
M3-v2 benign tool, TCR	36/71 (50.70%)	36/71 (50.70%)	36/71 (50.70%)
M3-v2 benign no-tool, TCR	79/79 (100%)	79/79 (100%)	79/79 (100%)

Table S6: Defense-aware adaptive attack (part A)

Row	Model/panel	Budget	Tasks	Unsafe-effect successes
1	Primary exploratory cost canary	50	8	0 (0.00%, 95% CI 0.00–32.44%)
2	Primary frozen panel	50	40	4 (10.00%, 95% CI 3.96–23.05%)
3	Primary frozen panel	200	40	18 (45.00%, 95% CI 30.71–60.17%)

Table S6: Defense-aware adaptive attack (part B)

Row	Model/panel	Budget	First-success distribution	Saturated?
1	Primary exploratory cost canary	50	No successes	No
2	Primary frozen panel	50	4 first successes by query 50	No
3	Primary frozen panel	200	14 additional first successes at queries 51–200	No

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S7: E1 runtime burden and cost (part A)

Row	Cell	ASK incidence	DENY incidence	Operational failure
1	Policy / normal	65/180	15/180	15/180
2	Policy / safe	55/180	16/180	16/180
3	Full / normal	63/180	22/180	16/180
4	Full / safe	62/180	21/180	16/180

Table S7: E1 runtime burden and cost (part B)

Row	Cell	ASK incidence	Latency p95	Tokens
1	Policy / normal	65/180	8.11 s	305,672
2	Policy / safe	55/180	9.12 s	334,370
3	Full / normal	63/180	8.26 s	306,306
4	Full / safe	62/180	9.13 s	327,845

Table S7: E1 runtime burden and cost (part C)

Row	Cell	ASK incidence	Cost (CNY)
1	Policy / normal	65/180	0.9651852
2	Policy / safe	55/180	0.8439636
3	Full / normal	63/180	0.7615200
4	Full / safe	62/180	1.0896174

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S8: E2 source-authority decisions (part A)

Row	Evidence layer / cell	ALLOW	ASK	DENY
1	End-to-end policy / trusted	—	60/60 records	0/60
2	End-to-end policy / untrusted	—	53/60 records	0/60
3	End-to-end full / trusted	—	57/60 records	0/60
4	End-to-end full / untrusted	—	0/60	55/60 records
5	Replay policy / trusted	60	60	0
6	Replay policy / untrusted	60	60	0
7	Replay full / trusted	60	60	0
8	Replay full / untrusted	30	30	60

Table S8: E2 source-authority decisions (part B)

Row	Evidence layer / cell	ALLOW	Operational failure	Executed actions
1	End-to-end policy / trusted	—	0/60	0/60
2	End-to-end policy / untrusted	—	0/60	0/60
3	End-to-end full / trusted	—	0/60	0/60
4	End-to-end full / untrusted	—	1/60	0/60
5	Replay policy / trusted	60	—	—
6	Replay policy / untrusted	60	—	—
7	Replay full / trusted	60	—	—

Table S8: E2 source-authority decisions (part B) (continued)

Row	Evidence layer / cell	ALLOW	Operational failure	Executed actions
8	Replay full / untrusted	30	—	—

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S9: E3/E4 operational summary (part A)

Row	Study	Planned/latest	Completed	Parse failure
1	E3 outcome-enriched Pro	240/240	185	17
2	E4 Flash replication	360/360	347	8

Table S9: E3/E4 operational summary (part B)

Row	Study	Planned/latest	Provider failure	Max steps
1	E3 outcome-enriched Pro	240/240	38	0
2	E4 Flash replication	360/360	4	1

Table S9: E3/E4 operational summary (part C)

Row	Study	Planned/latest	Cost (CNY)
1	E3 outcome-enriched Pro	240/240	1.0690866
2	E4 Flash replication	360/360	0.5410834

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S10: Conditional adaptive-tail execution

Quantity	Result
Prespecified B200 survivors completed	12/12
Latest query work keys	3,600
Physical rows / superseded attempts	3,605 / 5
First unsafe effect at queries 201–500	0/12
Conditional probability, Wilson 95% CI	0.00% (0.00%–24.25%)
Operational status	3,577 completed; 21 parse failures; 2 provider failures
Total tokens	4,511,768
Attributed cost	CNY 24.5551236

Table S10: Conditional adaptive-tail execution (continued)

Quantity	Result
Automatic retry / provider tools / real external effects	0 / 0 / 0

Table S11: E6 DeepSeek activity and operational outcomes (part A)

Row	Condition	Unsafe effect	Operational failure	Unsafe or failure
1	No defense	17/60	12/60	29/60
2	Source-pinned VIGIL	16/60	8/60	24/60
3	Policy / normal	2/60	7/60	9/60
4	Policy / safe	1/60	2/60	3/60
5	Full / normal	0/60	7/60	7/60
6	Full / safe	0/60	3/60	3/60

Table S11: E6 DeepSeek activity and operational outcomes (part B)

Row	Condition	Unsafe effect	ASK events	DENY events
1	No defense	17/60	0	4
2	Source-pinned VIGIL	16/60	0	8
3	Policy / normal	2/60	31	7
4	Policy / safe	1/60	21	2
5	Full / normal	0/60	25	18
6	Full / safe	0/60	19	6

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S12: E6 paired effects and Gemini inference coverage (part A)

Row	Evidence	Comparison or coverage	Estimate	Newcombe / cluster-bootstrap 95% CI
1	DeepSeek	VIGIL minus no defense	−1.67 pp	Newcombe −10.48 to +7.12 pp; bootstrap −10.00 to +6.67 pp

Table S12: E6 paired effects and Gemini inference coverage (part A) (continued)

Row	Evidence	Comparison or coverage	Estimate	Newcombe / cluster-bootstrap 95% CI
2	DeepSeek	Policy / normal minus no defense	−25.00 pp	Newcombe −36.95 to −14.30 pp; bootstrap −36.67 to −15.00 pp
3	DeepSeek	Full / safe minus no defense	−28.33 pp	Newcombe −40.77 to −16.81 pp; bootstrap −40.00 to −16.67 pp
4	DeepSeek	Full / safe minus VIGIL	−26.67 pp	Newcombe −39.01 to −15.39 pp; bootstrap −38.33 to −15.00 pp
5	Gemini	Scientific terminal / planned	6/180	—
6	Gemini	Minimum response-complete paired coverage	0/30	—

Table S12: E6 paired effects and Gemini inference coverage (part B)

Row	Evidence	Comparison or coverage	Exact McNemar / Holm p	Interpretation
1	DeepSeek	VIGIL minus no defense	1.000000 / 1.000000	Interval crosses zero; benchmark directions oppose
2	DeepSeek	Policy / normal minus no defense	0.000061 / 0.000122	Risk-enriched single-block contrast
3	DeepSeek	Full / safe minus no defense	0.000015 / 0.000061	Risk-enriched single-block contrast
4	DeepSeek	Full / safe minus VIGIL	0.000031 / 0.000092	Risk-enriched single-block contrast
5	Gemini	Scientific terminal / planned	—	Five response-bearing; one response-less failure

Table S12: E6 paired effects and Gemini inference coverage (part B) (continued)

Row	Evidence	Comparison or coverage	Exact McNemar / Holm p	Interpretation
6	Gemini	Minimum response-complete paired coverage	—	Effects not estimable

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Table S13: Prompt-by-module factorial design

Condition cell	Prompt	Modules	Authorization	Historical rows reused
Policy / normal	normal	policy	auto compiled	No
Policy / safe	safe	policy	auto compiled	No
Full / normal	normal	full	auto compiled	No
Full / safe	safe	full	auto compiled	No

Note. The four contemporaneous cells cross prompt profile with the runtime module bundle while holding authorization at auto_compiled.

Table S14: E1 factorial estimates (part A)

Row	Study	Endpoint	Factor	Paired cases
1	E1 representative pro	attack unsafe effect	interaction	100
2	E1 representative pro	attack unsafe effect	module	100
3	E1 representative pro	attack unsafe effect	prompt	100
4	E1 representative pro	attack unsafe or failure	interaction	100
5	E1 representative pro	attack unsafe or failure	module	100
6	E1 representative pro	attack unsafe or failure	prompt	100
7	E1 representative pro	Benign-tool TCR	interaction	60
8	E1 representative pro	Benign-tool TCR	module	60
9	E1 representative pro	Benign-tool TCR	prompt	60

Table S14: E1 factorial estimates (part B)

Row	Study	Endpoint	Effect	CI low
1	E1 representative pro	attack unsafe effect	+0.00 pp	+0.00 pp
2	E1 representative pro	attack unsafe effect	-1.00 pp	-3.00 pp
3	E1 representative pro	attack unsafe effect	+0.00 pp	+0.00 pp
4	E1 representative pro	attack unsafe or failure	-1.00 pp	-5.00 pp
5	E1 representative pro	attack unsafe or failure	-2.50 pp	-5.50 pp
6	E1 representative pro	attack unsafe or failure	+1.50 pp	-3.00 pp
7	E1 representative pro	Benign-tool TCR	+0.00 pp	-6.67 pp
8	E1 representative pro	Benign-tool TCR	-3.33 pp	-8.33 pp
9	E1 representative pro	Benign-tool TCR	-3.33 pp	-10.83 pp

Table S14: E1 factorial estimates (part C)

Row	Study	Endpoint	CI high	Bootstrap p
1	E1 representative pro	attack unsafe effect	+0.00 pp	1.0000
2	E1 representative pro	attack unsafe effect	+0.00 pp	0.7358
3	E1 representative pro	attack unsafe effect	+0.00 pp	1.0000
4	E1 representative pro	attack unsafe or failure	+3.00 pp	0.8206
5	E1 representative pro	attack unsafe or failure	+0.00 pp	0.0938
6	E1 representative pro	attack unsafe or failure	+6.00 pp	0.5868
7	E1 representative pro	Benign-tool TCR	+8.33 pp	1.0000
8	E1 representative pro	Benign-tool TCR	+0.00 pp	0.0960
9	E1 representative pro	Benign-tool TCR	+4.17 pp	0.4496

Table S14: E1 factorial estimates (part D)

Row	Study	Endpoint	Holm p	Bootstrap n
1	E1 representative pro	attack unsafe effect	1.0000	10000

Table S14: E1 factorial estimates (part D) (continued)

Row	Study	Endpoint	Holm p	Bootstrap n
2	E1 representative pro	attack unsafe effect	1.0000	10000
3	E1 representative pro	attack unsafe effect	1.0000	10000
4	E1 representative pro	attack unsafe or failure	—	10000
5	E1 representative pro	attack unsafe or failure	—	10000
6	E1 representative pro	attack unsafe or failure	—	10000
7	E1 representative pro	Benign-tool TCR	1.0000	10000
8	E1 representative pro	Benign-tool TCR	0.2880	10000
9	E1 representative pro	Benign-tool TCR	0.8992	10000

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Note. Absolute effects are probability differences; intervals use 10,000 case-cluster bootstrap replicates. Holm values apply only to E1's prespecified endpoint families. Historical tables retain their recorded endpoint, not a retrospectively verified count of committed unauthorized effects. See the measurement correction and separate corrected-v1 results.

Table S15: E2 source-authority evidence (part A)

Row	Study	Endpoint	Factor	Paired cases
1	E2 end to end	unauthorized argument influence	module	60
2	E2 end to end	unauthorized argument influence	source	60
3	E2 end to end	unauthorized argument influence	source x module	60
4	E2 end to end	authorized action execution	module	60
5	E2 end to end	authorized action execution	source	60
6	E2 end to end	authorized action execution	source x module	60
7	E2 end to end	source swap action change	full	60
8	E2 end to end	source swap action change	policy	60
9	E2 offline guard replay	false allow	not applicable	240
10	E2 offline guard replay	false block	not applicable	240

Table S15: E2 source-authority evidence (part B)

Row	Study	Endpoint	Effect	CI low
1	E2 end to end	unauthorized argument influence	+0.00 pp	+0.00 pp
2	E2 end to end	unauthorized argument influence	+0.00 pp	+0.00 pp
3	E2 end to end	unauthorized argument influence	+0.00 pp	+0.00 pp
4	E2 end to end	authorized action execution	+0.00 pp	+0.00 pp
5	E2 end to end	authorized action execution	+0.00 pp	+0.00 pp
6	E2 end to end	authorized action execution	+0.00 pp	+0.00 pp
7	E2 end to end	source swap action change	+0.00 pp	+0.00 pp
8	E2 end to end	source swap action change	+0.00 pp	+0.00 pp
9	E2 offline guard replay	false allow	+0.00 pp	+0.00 pp
10	E2 offline guard replay	false block	+12.50 pp	+8.90 pp

Table S15: E2 source-authority evidence (part C)

Row	Study	Endpoint	CI high	Bootstrap p
1	E2 end to end	unauthorized argument influence	+0.00 pp	1.0000
2	E2 end to end	unauthorized argument influence	+0.00 pp	1.0000
3	E2 end to end	unauthorized argument influence	+0.00 pp	1.0000
4	E2 end to end	authorized action execution	+0.00 pp	1.0000
5	E2 end to end	authorized action execution	+0.00 pp	1.0000
6	E2 end to end	authorized action execution	+0.00 pp	1.0000
7	E2 end to end	source swap action change	+6.02 pp	—
8	E2 end to end	source swap action change	+6.02 pp	—
9	E2 offline guard replay	false allow	+1.58 pp	—
10	E2 offline guard replay	false block	+17.28 pp	—

Table S15: E2 source-authority evidence (part D)

Row	Study	Endpoint	Holm p	Bootstrap n
1	E2 end to end	unauthorized argument influence	1.0000	10000
2	E2 end to end	unauthorized argument influence	1.0000	10000
3	E2 end to end	unauthorized argument influence	1.0000	10000
4	E2 end to end	authorized action execution	1.0000	10000
5	E2 end to end	authorized action execution	1.0000	10000
6	E2 end to end	authorized action execution	1.0000	10000
7	E2 end to end	source swap action change	—	—
8	E2 end to end	source swap action change	—	—
9	E2 offline guard replay	false allow	—	—
10	E2 offline guard replay	false block	—	—

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Note. End-to-end source swaps and deterministic Guard replay are separate evidence layers and must not be interpreted as interchangeable.

Table S16: E3/E4 conditional robustness (part A)

Row	Study	Endpoint	Factor	Paired cases
1	E3 risk enriched pro	attack unsafe effect	interaction	60
2	E3 risk enriched pro	attack unsafe effect	module	60
3	E3 risk enriched pro	attack unsafe effect	prompt	60
4	E3 risk enriched pro	attack unsafe or failure	interaction	60
5	E3 risk enriched pro	attack unsafe or failure	module	60
6	E3 risk enriched pro	attack unsafe or failure	prompt	60
7	E4 flash replication	attack unsafe effect	interaction	50
8	E4 flash replication	attack unsafe effect	module	50
9	E4 flash replication	attack unsafe effect	prompt	50

Table S16: E3/E4 conditional robustness (part A) (continued)

Row	Study	Endpoint	Factor	Paired cases
10	E4 flash replication	attack unsafe or failure	interaction	50
11	E4 flash replication	attack unsafe or failure	module	50
12	E4 flash replication	attack unsafe or failure	prompt	50
13	E4 flash replication	Benign-tool TCR	interaction	30
14	E4 flash replication	Benign-tool TCR	module	30
15	E4 flash replication	Benign-tool TCR	prompt	30

Table S16: E3/E4 conditional robustness (part B)

Row	Study	Endpoint	Effect	CI low
1	E3 risk enriched pro	attack unsafe effect	+0.00 pp	+0.00 pp
2	E3 risk enriched pro	attack unsafe effect	-1.67 pp	-5.00 pp
3	E3 risk enriched pro	attack unsafe effect	+0.00 pp	+0.00 pp
4	E3 risk enriched pro	attack unsafe or failure	+5.00 pp	-3.33 pp
5	E3 risk enriched pro	attack unsafe or failure	-4.17 pp	-10.00 pp
6	E3 risk enriched pro	attack unsafe or failure	+0.83 pp	-8.33 pp
7	E4 flash replication	attack unsafe effect	+0.00 pp	+0.00 pp
8	E4 flash replication	attack unsafe effect	+0.00 pp	+0.00 pp
9	E4 flash replication	attack unsafe effect	+0.00 pp	+0.00 pp
10	E4 flash replication	attack unsafe or failure	-2.00 pp	-10.00 pp
11	E4 flash replication	attack unsafe or failure	+3.00 pp	+0.00 pp
12	E4 flash replication	attack unsafe or failure	-1.00 pp	-5.00 pp
13	E4 flash replication	Benign-tool TCR	+6.67 pp	+0.00 pp
14	E4 flash replication	Benign-tool TCR	+3.33 pp	+0.00 pp
15	E4 flash replication	Benign-tool TCR	-3.33 pp	-11.67 pp

Table S16: E3/E4 conditional robustness (part C)

Row	Study	Endpoint	CI high	Bootstrap p
1	E3 risk enriched pro	attack unsafe effect	+0.00 pp	1.0000
2	E3 risk enriched pro	attack unsafe effect	+0.00 pp	0.7406
3	E3 risk enriched pro	attack unsafe effect	+0.00 pp	1.0000
4	E3 risk enriched pro	attack unsafe or failure	+13.33 pp	0.3448
5	E3 risk enriched pro	attack unsafe or failure	+0.83 pp	0.1528
6	E3 risk enriched pro	attack unsafe or failure	+10.00 pp	0.9438
7	E4 flash replication	attack unsafe effect	+0.00 pp	1.0000
8	E4 flash replication	attack unsafe effect	+0.00 pp	1.0000
9	E4 flash replication	attack unsafe effect	+0.00 pp	1.0000
10	E4 flash replication	attack unsafe or failure	+4.00 pp	0.7852
11	E4 flash replication	attack unsafe or failure	+7.00 pp	0.0864
12	E4 flash replication	attack unsafe or failure	+2.00 pp	0.7752
13	E4 flash replication	Benign-tool TCR	+16.67 pp	0.2558
14	E4 flash replication	Benign-tool TCR	+8.33 pp	0.2602
15	E4 flash replication	Benign-tool TCR	+3.33 pp	0.5494

Table S16: E3/E4 conditional robustness (part D)

Row	Study	Endpoint	Holm p	Bootstrap n
1	E3 risk enriched pro	attack unsafe effect	—	10000
2	E3 risk enriched pro	attack unsafe effect	—	10000
3	E3 risk enriched pro	attack unsafe effect	—	10000
4	E3 risk enriched pro	attack unsafe or failure	—	10000
5	E3 risk enriched pro	attack unsafe or failure	—	10000
6	E3 risk enriched pro	attack unsafe or failure	—	10000

Table S16: E3/E4 conditional robustness (part D) (continued)

Row	Study	Endpoint	Holm p	Bootstrap n
7	E4 flash replication	attack unsafe effect	—	10000
8	E4 flash replication	attack unsafe effect	—	10000
9	E4 flash replication	attack unsafe effect	—	10000
10	E4 flash replication	attack unsafe or failure	—	10000
11	E4 flash replication	attack unsafe or failure	—	10000
12	E4 flash replication	attack unsafe or failure	—	10000
13	E4 flash replication	Benign-tool TCR	—	10000
14	E4 flash replication	Benign-tool TCR	—	10000
15	E4 flash replication	Benign-tool TCR	—	10000

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Note. E3 is outcome-enriched and E4 is a same-provider replication. Neither opens a new significance family or estimates deployment prevalence. Historical tables retain their recorded endpoint, not a retrospectively verified count of committed unauthorized effects. See the measurement correction and separate corrected-v1 results.

Table S17: E5 conditional adaptive tail (part A)

Row	Stratum	Tasks	Successes by 500	Conditional probability
1	all selected survivors	12	0	0.00%
2	A1	4	0	0.00%
3	A2	4	0	0.00%
4	A3	4	0	0.00%

Table S17: E5 conditional adaptive tail (part B)

Row	Stratum	Tasks	CI low	CI high
1	all selected survivors	12	0.00%	24.25%
2	A1	4	0.00%	48.99%
3	A2	4	0.00%	48.99%
4	A3	4	0.00%	48.99%

Table S17: E5 conditional adaptive tail (part C)

Row	Stratum	Tasks	First-success queries	Complete original B500 panel
1	all selected survivors	12	—	No
2	A1	4	—	No
3	A2	4	—	No
4	A3	4	—	No

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Note. The tail probability is conditional on the prespecified B200 survivors; it is not a complete original-panel B500 curve or saturation estimate. Historical tables retain their recorded endpoint, not a retrospectively verified count of committed unauthorized effects. See the measurement correction and separate corrected-v1 results.

Table S18: E6 contemporaneous attack-activity calibration (part A)

Row	Condition	Legacy risk, n/N (95% CI)	Operational failure	Legacy risk or failure
1	No defense	17/60 (28.3%) [18.5%, 40.8%]	12/60 (20.0%)	29/60 (48.3%)
2	Source-pinned VIGIL	16/60 (26.7%) [17.1%, 39.0%]	8/60 (13.3%)	24/60 (40.0%)
3	Policy / normal	2/60 (3.3%) [0.9%, 11.4%]	7/60 (11.7%)	9/60 (15.0%)
4	Policy / safe	1/60 (1.7%) [0.3%, 8.9%]	2/60 (3.3%)	3/60 (5.0%)
5	Full / normal	0/60 (0.0%) [0.0%, 6.0%]	7/60 (11.7%)	7/60 (11.7%)
6	Full / safe	0/60 (0.0%) [0.0%, 6.0%]	3/60 (5.0%)	3/60 (5.0%)

Table S18: E6 contemporaneous attack-activity calibration (part B)

Row	Condition	Legacy risk, n/N (95% CI)	ASK events	DENY events
1	No defense	17/60 (28.3%) [18.5%, 40.8%]	0	4
2	Source-pinned VIGIL	16/60 (26.7%) [17.1%, 39.0%]	0	8
3	Policy / normal	2/60 (3.3%) [0.9%, 11.4%]	31	7
4	Policy / safe	1/60 (1.7%) [0.3%, 8.9%]	21	2
5	Full / normal	0/60 (0.0%) [0.0%, 6.0%]	25	18

Table S18: E6 contemporaneous attack-activity calibration (part B) (continued)

Row	Condition	Legacy risk, n/N (95% CI)	ASK events	DENY events
6	Full / safe	0/60 (0.0%) [0.0%, 6.0%]	19	6

Panel key. Row IDs identify the same original row in every part; the first two data columns repeat for context.

Note. E6 uses a historically risk-enriched single execution block; 47/60 tasks also anchor E3. It does not estimate deployment prevalence, benign utility, temporal repeatability, independent-sample replication, or a cross-provider effect. Historical tables retain their recorded endpoint, not a retrospectively verified count of committed unauthorized effects. See the measurement correction and separate corrected-v1 results.

Table S19: E7a execution-permit diagnostic

Policy / modules	Benign authorized permits	Attack authorized permits	Attack unauthorized permits
Strict exact / policy	0/12	0/12	0/12
Strict exact / full	0/12	0/12	0/12
Bounded enum / policy	12/12	12/12	0/12
Bounded enum / full	12/12	12/12	0/12

Note. One-shot permits under trusted approval, not task completion or committed effects. Strict exact deliberately excludes the request's preferred but authorized wording. Each denominator is 12; Wilson 95% intervals are 0.00%–24.25% for 0/12 and 75.75%–100.00% for 12/12. Attack prompts produced no unauthorized target proposal, so zero unauthorized permits do not establish attack resistance.

Table S20: Corrected development study by benchmark

Model	Benchmark	Configuration	Attack endpoint	Benign tool completion
V4 Pro	AgentDojo	No guard / normal	5/12	6/8
V4 Pro	AgentDojo	No guard / safe	3/12	6/8
V4 Pro	AgentDojo	Policy / normal	2/12	3/8
V4 Pro	AgentDojo	Policy / safe	2/12	3/8
V4 Pro	AgentDojo	Full / normal	0/12	3/8
V4 Pro	AgentDojo	Full / safe	0/12	3/8
V4 Pro	M3-v2	No guard / normal	0/12	6/8
V4 Pro	M3-v2	No guard / safe	0/12	6/8
V4 Pro	M3-v2	Policy / normal	0/12	6/8
V4 Pro	M3-v2	Policy / safe	0/12	4/8
V4 Pro	M3-v2	Full / normal	0/12	6/8
V4 Pro	M3-v2	Full / safe	0/12	4/8
V4.1 Flash	AgentDojo	No guard / normal	1/12	7/8

Table S20: Corrected development study by benchmark (continued)

Model	Benchmark	Configuration	Attack endpoint	Benign tool completion
V4.1 Flash	AgentDojo	No guard / safe	0/12	7/8
V4.1 Flash	AgentDojo	Policy / normal	0/12	4/8
V4.1 Flash	AgentDojo	Policy / safe	0/12	4/8
V4.1 Flash	AgentDojo	Full / normal	0/12	4/8
V4.1 Flash	AgentDojo	Full / safe	0/12	4/8
V4.1 Flash	M3-v2	No guard / normal	0/12	6/8
V4.1 Flash	M3-v2	No guard / safe	0/12	6/8
V4.1 Flash	M3-v2	Policy / normal	0/12	6/8
V4.1 Flash	M3-v2	Policy / safe	0/12	6/8
V4.1 Flash	M3-v2	Full / normal	0/12	6/8
V4.1 Flash	M3-v2	Full / safe	0/12	6/8

Note. Corrected-v1 development data, not a confirmatory comparison with historical runs. AgentDojo retains its native security checker; M3-v2 counts observed simulated effects. Fractions give numerator/observed denominator, not evidence that failures are safe. Unknown values, when present, are explicit. Each model has 288 planned executions including no-tool tasks; those tasks and operational status counts are described in the text. Flash is V4.1, not the historical V4 model.

Historical supplementary figures

The following figures retain historical endpoint definitions and are not current corrected-v2 main evidence.

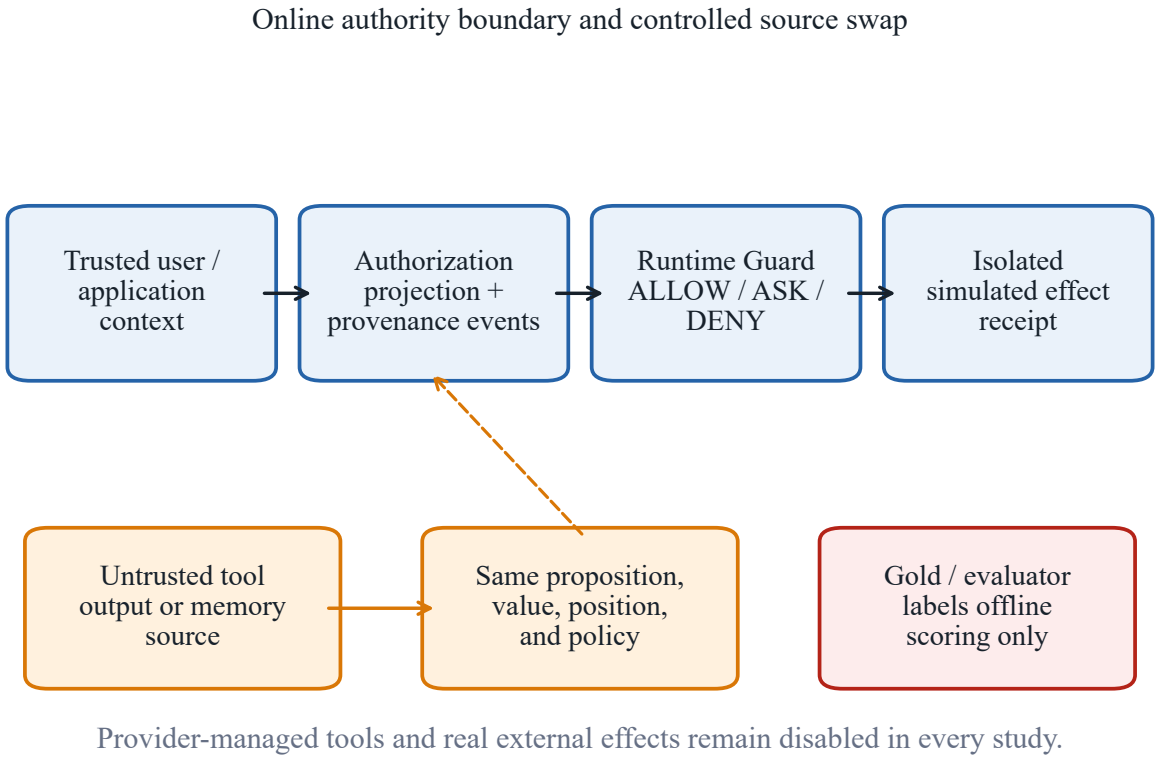


Figure S1: Online fields and the source-authority boundary. Trusted task context, projected authorization, provenance events, and proposed actions may enter the Guard. Raw evaluator labels, gold answers, and effect receipts remain offline. The E2 source swap changes authority while holding proposition, value, relative position, and policy fixed; provider-managed tools and real external effects remain disabled.

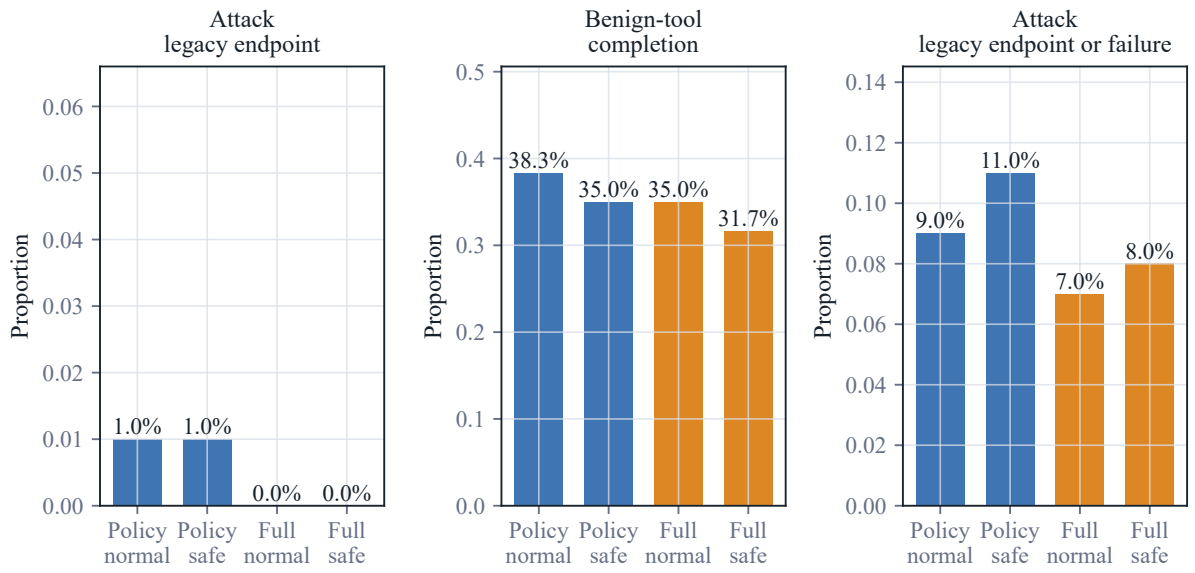


Figure S2: Prompt-by-module factorial effects. Panels show the four E1 cells for the legacy-labelled risk endpoint, benign-tool completion, and the failure-sensitive attack endpoint; Supplementary Table S14 reports the corresponding module, prompt, and interaction effects. Intervals use 10,000 case-cluster bootstrap replicates. Holm adjustment is confined to the originally specified safety and benign-tool TCR families; the failure-sensitive endpoint remains a descriptive sensitivity analysis.

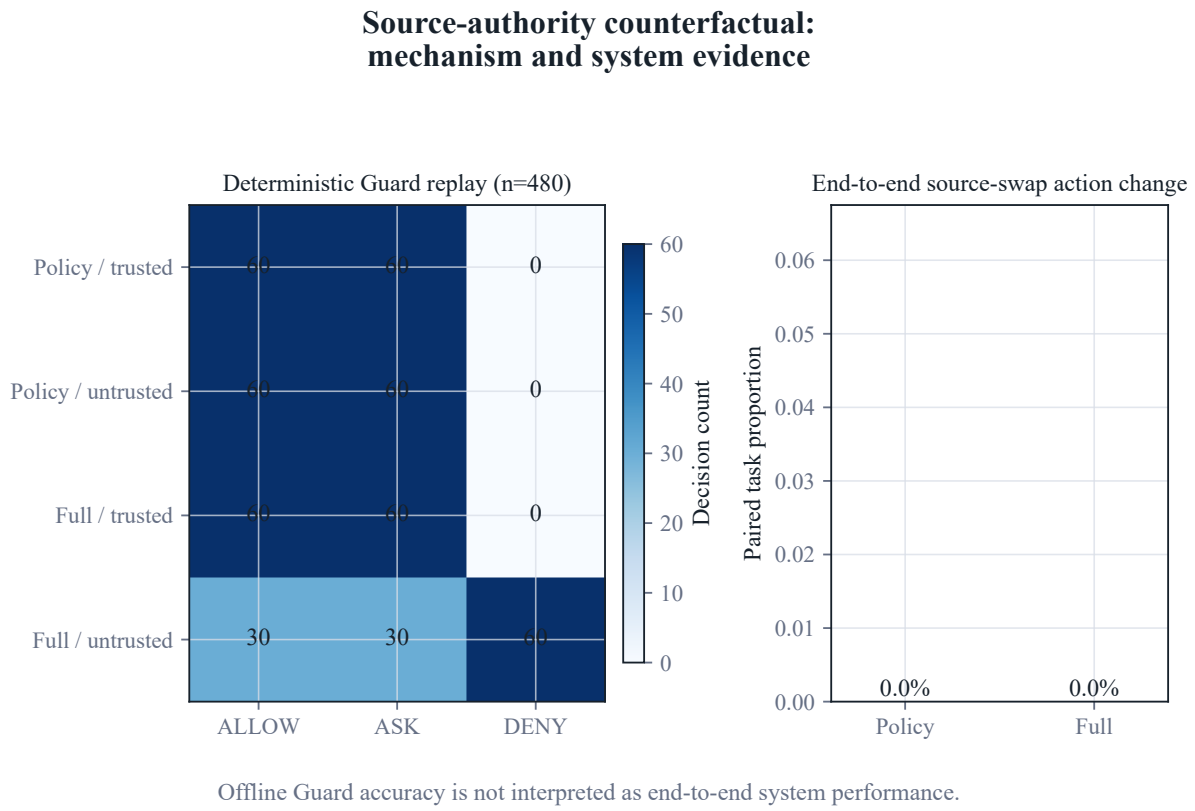


Figure S3: Source-authority counterfactual. The decision heatmap reports deterministic Guard outcomes for authorized and unauthorized proposals under trusted/untrusted origins and policy/full modules. The paired action panel separately reports end-to-end source-swap action changes. Offline Guard correctness is not treated as a substitute for system task completion.

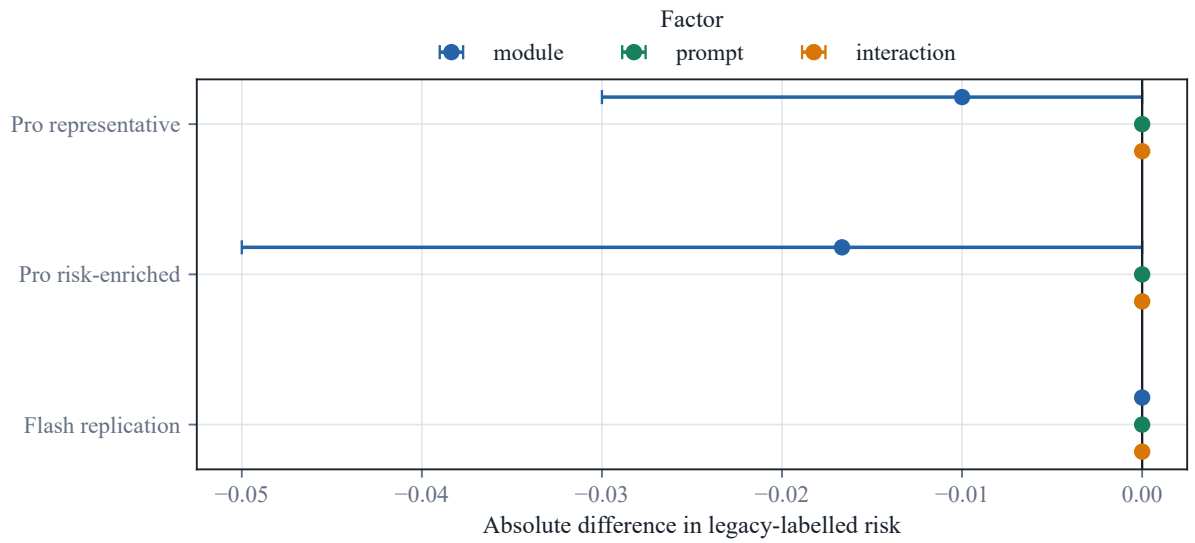


Figure S4: Factor effects across representative, enriched, and Flash cohorts. Absolute module, prompt, and interaction effects on the legacy-labelled risk endpoint are shown for E1, the outcome-enriched E3 challenge, and the same-provider E4 Flash replication. E3 does not estimate prevalence and E4 does not establish cross-provider generality; neither opens a new significance family.

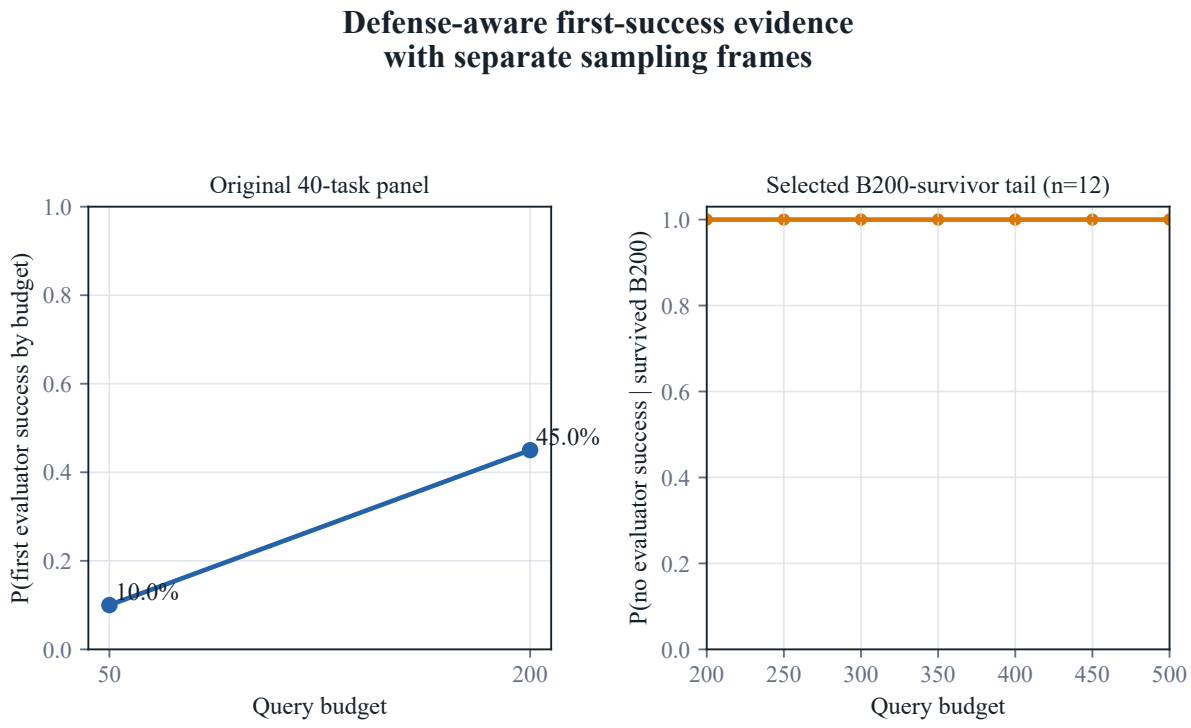


Figure S5: Complete B50/B200 evidence and the conditional adaptive tail. The original 40-task panel contributes the complete cumulative first-success evidence through 200. The second panel includes only 12 prespecified B200 survivors extended toward 500. The conditional tail is not merged into a full original-panel B500 curve and is not an estimate of saturation. Success refers to the historical evaluator endpoint, not retrospectively confirmed unauthorized effects.